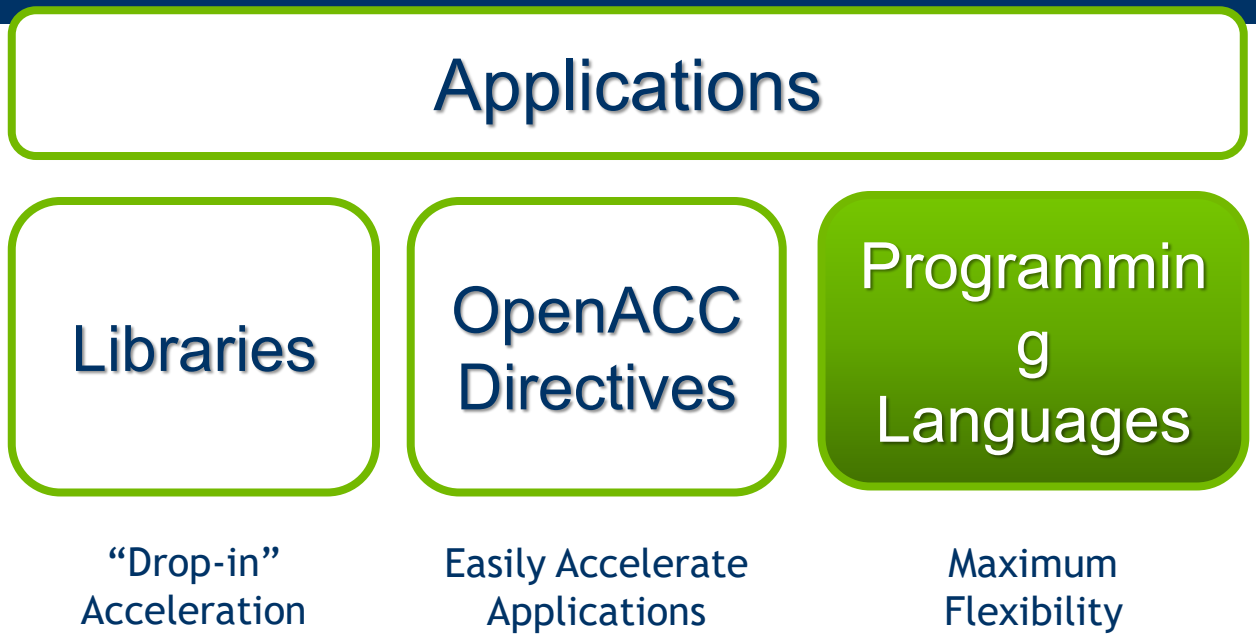
Parallel & Distributed Computing
Lecture 20: Programming With CUDA

Farhad M. Riaz

Farhad.Muhammad@numl.edu.pk

Department of Computer Science
NUML, Islamabad

3 Ways to Accelerate Applications



What is CUDA?

- Compute Unified Device Architecture (CUDA)
  - A powerful software platform that helps computer program run faster.
  - Often used to solve performance-intensive problems, such as:
    - **Crypto mining**
    - **Video rendering**
    - **Machine learning**
    - **Etc**
  - CUDA is also embedded in hardware. Specifically: NVIDIA graphic cards

What is CUDA?

- CUDA Architecture
  - Expose GPU parallelism for general-purpose computing
  - Retain performance
- CUDA C/C++
  - Based on industry-standard C/C++
  - Small set of extensions to enable heterogeneous programming
  - Straightforward APIs to manage devices, memory etc.
- This session introduces CUDA C/C++

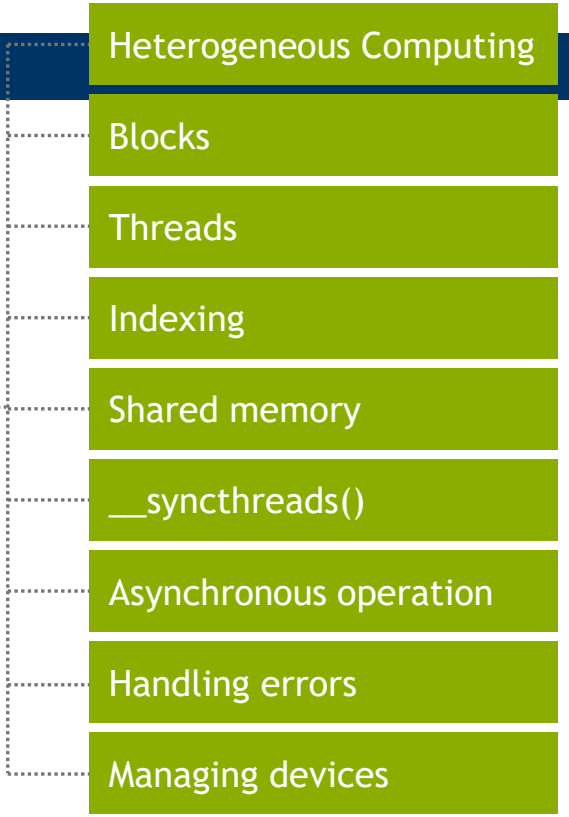
Introduction to CUDA C/C++

- What will you learn in this session?
  - Start from “Hello World!”
  - Write and launch CUDA C/C++ kernels
  - Manage GPU memory
  - Manage communication and synchronization

Prerequisites

- You (probably) need experience with C or C++
- You don’t need GPU experience
- You don’t need parallel programming experience
- You don’t need graphics experience

CONCEPTS


Heterogeneous Computing
Blocks
Threads
Indexing
Shared memory
__syncthreads()
Asynchronous operation
Handling errors
Managing devices

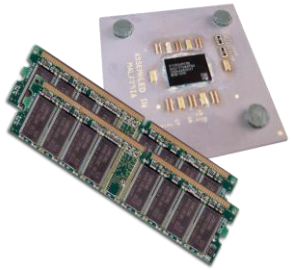
CONCEPTS

- Heterogeneous Computing
- Blocks
- Threads
- Indexing
- Shared memory
- __syncthreads()
- Asynchronous operation
- Handling errors
- Managing devices

HELLO WORLD!

Heterogeneous Computing

- Terminology:
  - *Host* The CPU and its memory (host memory)
  - *Device* The GPU and its memory (device memory)
  - **Host Memory**
    - the system main memory
  - **Device memory**
    - onboard memory on a GPU card
  - **kernel**
    - a GPU function launched by the host and executed on the device
  - **device function**
    - a GPU function executed on the device which can only be called from the device (i.e. from a kernel or another device function)



Host

Device

Heterogeneous Computing

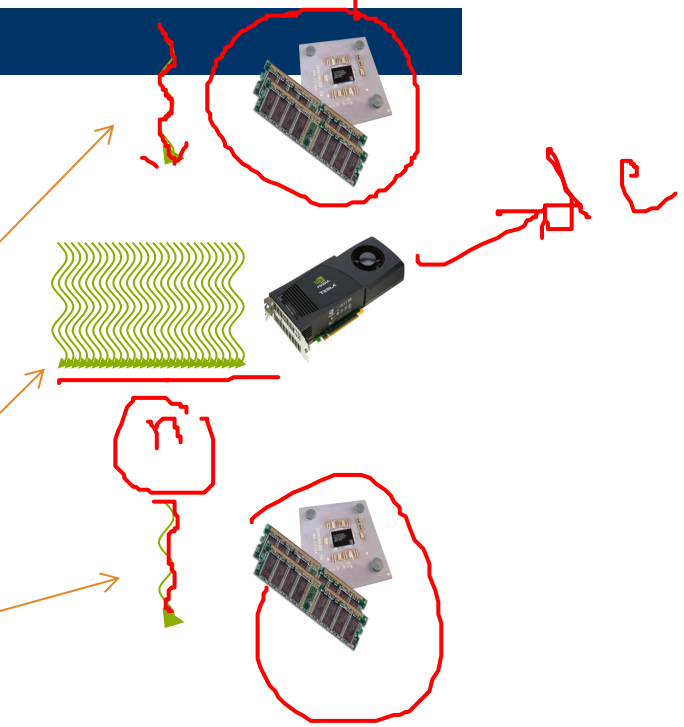
```
#include <iostream>
#include <algorithm>

using namespace std;

#define N          1024
#define RADIUS     3
#define BLOCK_SIZE 16

__global__ void stencil_1d(int *in, int *out) {
        __shared__ int temp[BLOCK_SIZE + 2 * RADIUS];
        int gindex = threadIdx.x + blockIdx.x * blockDim.x;
        int lindex = threadIdx.x + RADIUS;

        // Read input elements into shared memory
        temp[lindex] = in[gindex];
        if (threadIdx.x < RADIUS) {
                temp[lindex - RADIUS] = in[gindex - RADIUS];
                temp[lindex + BLOCK_SIZE] = in[gindex + BLOCK_SIZE];
        }

        // Synchronize (ensure all the data is available)
        __syncthreads();

        // Apply the stencil
        int result = 0;
        for (int offset = -RADIUS ; offset <= RADIUS ; offset++)
                result += temp[lindex + offset];

        // Store the result
        out[gindex] = result;
}

void fill_ints(int *x, int n) {
        fill_n(x, n, 1);
}

int main(void) {
        int *in, *out;          // host copies of a, b, c
        int *d_in, *d_out;      // device copies of a, b, c
        int size = (N + 2*RADIUS) * sizeof(int);

        // Alloc space for host copies and setup values
        in = (int *)malloc(size); fill_ints(in, N + 2*RADIUS);
        out = (int *)malloc(size); fill_ints(out, N + 2*RADIUS);

        // Alloc space for device copies
        cudaMalloc((void **)&d_in, size);
        cudaMalloc((void **)&d_out, size);

        // Copy to device
        cudaMemcpy(d_in, in, size, cudaMemcpyHostToDevice);
        cudaMemcpy(d_out, out, size, cudaMemcpyHostToDevice);

        // Launch stencil_1d() kernel on GPU
        stencil_1d<<<N/BLOCK_SIZE,BLOCK_SIZE>>>(d_in + RADIUS,
d_out + RADIUS);

        // Copy result back to host
        cudaMemcpy(out, d_out, size, cudaMemcpyDeviceToHost);

        // Cleanup
        free(in); free(out);
        cudaFree(d_in); cudaFree(d_out);
        return 0;
}
```

parallel fn

serial code

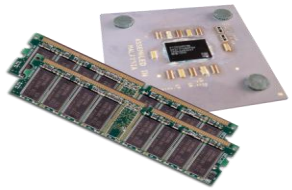
parallel code

serial code

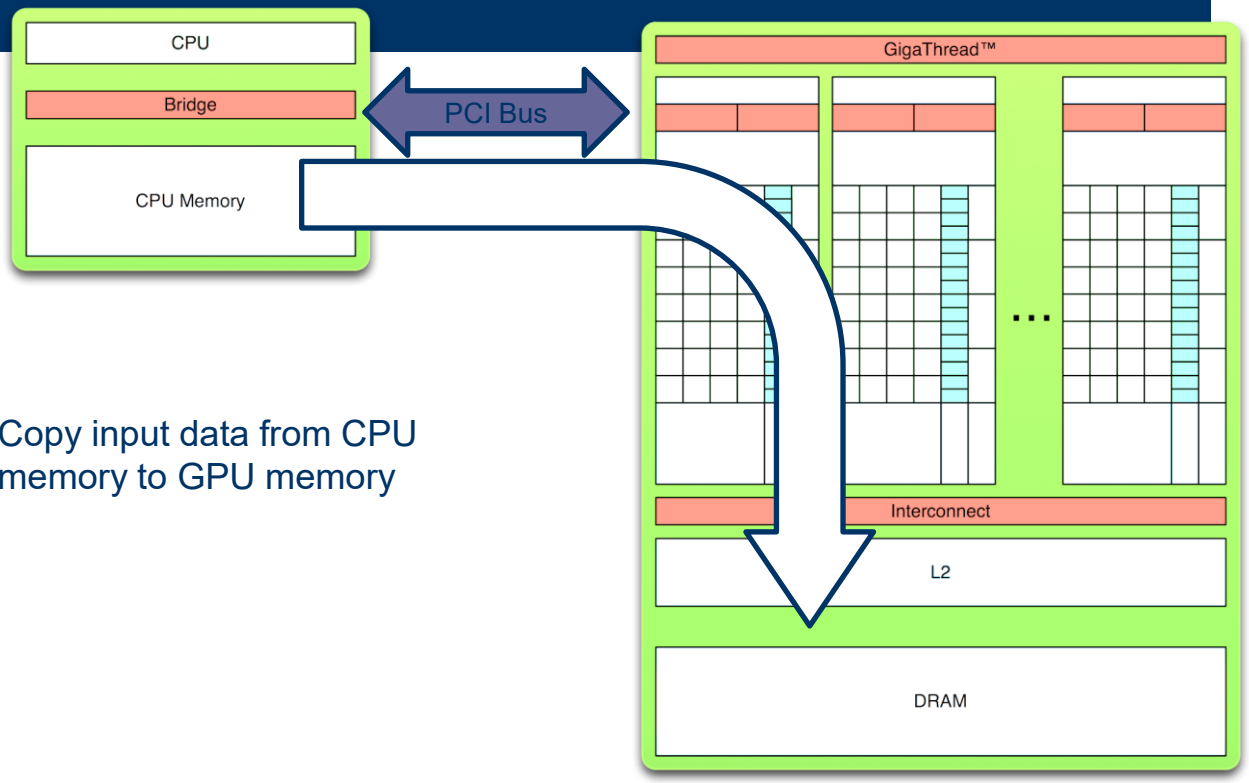


```
__global__
void saxpy(int n, float a, float *x, float *y)
{
  int i = blockIdx.x*blockDim.x + threadIdx.x;
  if (i < n) y[i] = a*x[i] + y[i];
}
```


```
x = (float*)malloc(N*sizeof(float));
 y = (float*)malloc(N*sizeof(float));

 cudaMalloc(&d_x, N*sizeof(float));
 cudaMalloc(&d_y, N*sizeof(float));
```

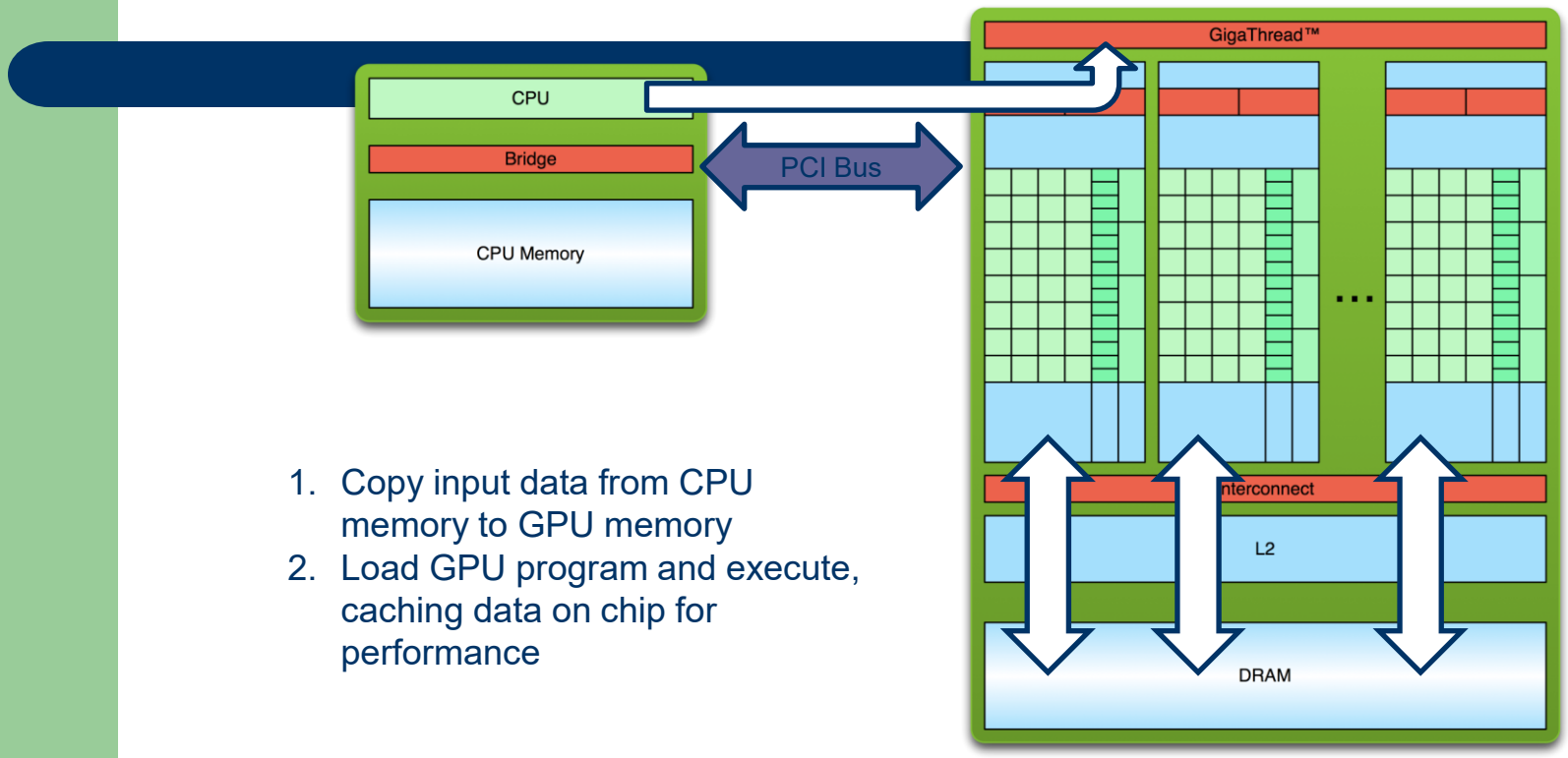


Simple Processing Flow



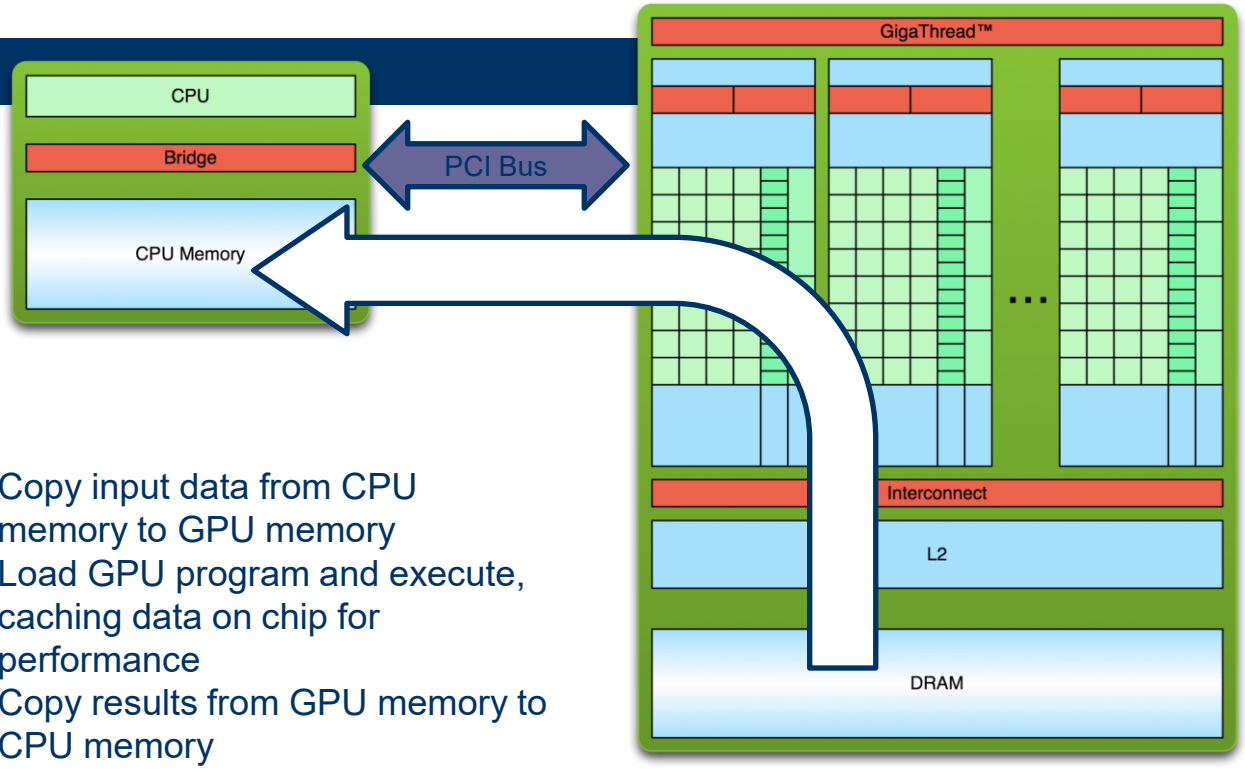
1. Copy input data from CPU memory to GPU memory

Simple Processing Flow



1. Copy input data from CPU memory to GPU memory
2. Load GPU program and execute, caching data on chip for performance

Simple Processing Flow



1. Copy input data from CPU memory to GPU memory
2. Load GPU program and execute, caching data on chip for performance
3. Copy results from GPU memory to CPU memory

C

```c
void c_hello(){
    printf("Hello World!\n");
}

int main() {
    c_hello();
    return 0;
}
```

CUDA

```cuda
__global__ void cuda_hello(){
    printf("Hello World from GPU!\n");
}

int main() {
    cuda_hello<<<1,1>>>();
    return 0;
}
```

Flow of CUDA Program

- Declare and allocate host and device memory.
- Initialize host data.
- Transfer data from the host to the device.
- Execute one or more kernels.
- Transfer results from the device to the host.

Hello World!

```
int main(void) {
printf("Hello World!\n");
        return 0;
}
```

- Standard C that runs on the host
- NVIDIA compiler (nvcc) can be used to compile programs with no *device* code

Output:

```
$ nvcc
hello_world.
cu
$ a.out
Hello World!
$
```

Hello World! with Device Code

```
__global__ void mykernel(void) {
}

int main(void) {
        mykernel<<<1,1>>>();
        printf("Hello World!\n");
        return 0;
}
```

- Two new syntactic elements…

Hello World! with Device Code

```
__global__ void mykernel(void) {
}
```

- CUDA C/C++ keyword `__global__` indicates a function that:
  - Runs on the device
  - Is called from host code

- `nvcc` separates source code into host and device components
  - Device functions (e.g. `mykernel()`) processed by NVIDIA compiler
  - Host functions (e.g. `main()`) processed by standard host compiler
    - **`gcc`, `cl.exe`**

Hello World! with Device Code

```
mykernel<<<1,1>>>();
```

- Triple angle brackets mark a call from *host* code to *device* code
  - Also called a “kernel launch”
  - We’ll return to the parameters (1,1) in a moment

- That’s all that is required to execute a function on the GPU!

Hello World! with Device Code

```
__global__ void mykernel(void){
        }

        int main(void) {
                mykernel<<<1,1>>>();
                printf("Hello
World!\n");
                return 0;
        }
```

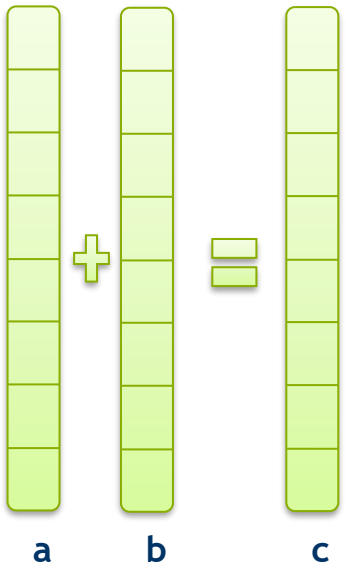
Output:

```
$ nvcc
hello.cu
$ a.out
Hello World!
$
```

- `mykernel()` does nothing, somewhat anticlimactic!

Parallel Programming in CUDA C/C++

- But wait… GPU computing is about massive parallelism!
- We need a more interesting example…
- We'll start by adding two integers and build up to vector addition



Addition on the Device

- A simple kernel to add two integers

```
__global__ void add(int *a, int *b, int *c) {
        *c = *a + *b;
}
```

- As before `__global__` is a CUDA C/C++ keyword meaning
  - `add()` will execute on the device
  - `add()` will be called from the host

Addition on the Device

- Note that we use pointers for the variables

```
__global__ void add(int *a, int *b, int *c) {
        *c = *a + *b;
}
```

- `add()` runs on the device, so `a`, `b` and `c` must point to device memory

- We need to allocate memory on the GPU

Memory Management

- Host and device memory are separate entities
  - *Device* pointers point to GPU memory
    May be passed to/from host code
    May *not* be dereferenced in host code
  - *Host* pointers point to CPU memory
    May be passed to/from device code
    May *not* be dereferenced in device code

- Simple CUDA API for handling device memory
  - `cudaMalloc()`, `cudaFree()`, `cudaMemcpy()`
  - Similar to the C equivalents `malloc()`, `free()`, `memcpy()`

Addition on the Device: `add()`

- Returning to our `add()` kernel

```
__global__ void add(int *a, int *b, int *c) {
        *c = *a + *b;
}
```

- Let's take a look at main()…

Addition on the Device: `main()`

```
int main(void) {
        int a, b, c;                    // host copies of
a, b, c
        int *d_a, *d_b, *d_c;           // device copies
of a, b, c

        int size = sizeof(int);

        // Allocate space for device copies of a, b, c
        cudaMalloc((void **)&d_a, size);
        cudaMalloc((void **)&d_b, size);
        cudaMalloc((void **)&d_c, size);

        // Setup input values
        a = 2;
        b = 7;
```

Addition on the Device: `main()`

```
    // Copy inputs to device
    cudaMemcpy(d_a, &a, size, cudaMemcpyHostToDevice);
    cudaMemcpy(d_b, &b, size, cudaMemcpyHostToDevice);

    // Launch add() kernel on GPU
    add<<<1,1>>>(d_a, d_b, d_c);

    // Copy result back to host
    cudaMemcpy(&c, d_c, size, cudaMemcpyDeviceToHost);

    // Cleanup
    cudaFree(d_a); cudaFree(d_b); cudaFree(d_c);
    return 0;
}
```

Another Example: Using Pytorch

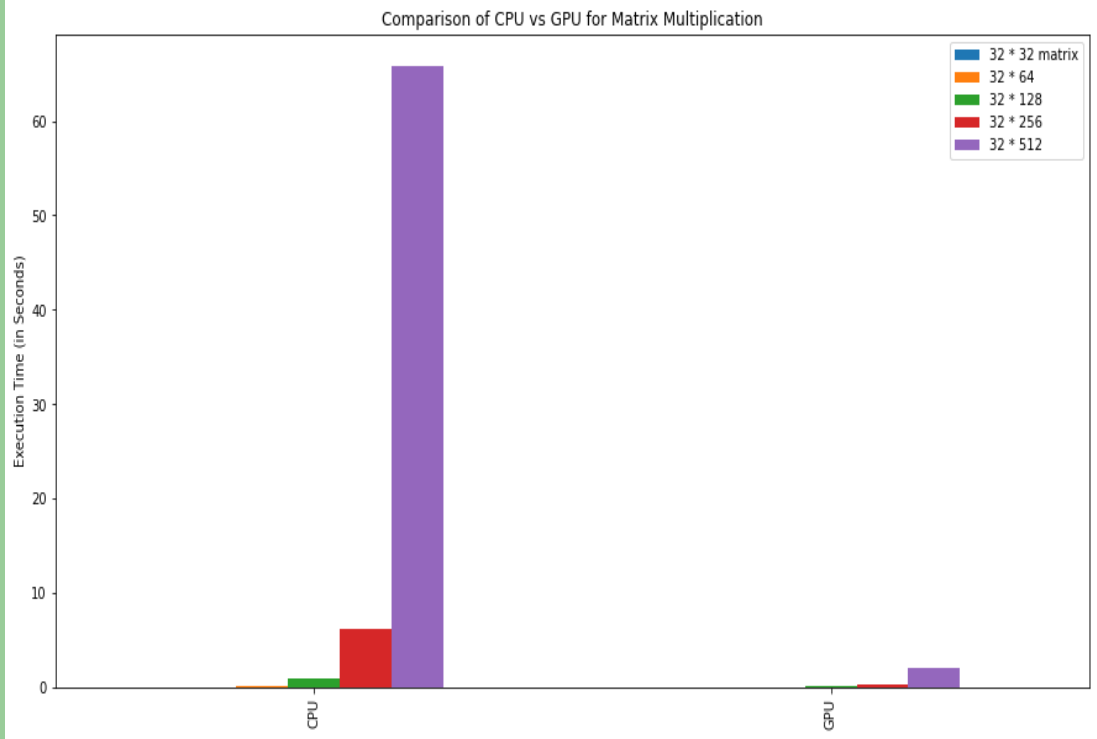
```
import torch

if torch.cuda.is_available():
  device = torch.device("cuda")
else:
  device = torch.device("cpu")

print("using", device, "device")
```

```
using cuda device
```

```
[2] import time

    matrix_size = 32*512

    x = torch.randn(matrix_size, matrix_size)
    y = torch.randn(matrix_size, matrix_size)

    print("******** CPU SPEED ********")
    start =time.time()
    result = torch.matmul(x,y)
    print(time.time() - start)
    print("verify device : ", result.device)


    x_gpu = x.to(device)
    y_gpu = y.to(device)
    torch.cuda.synchronize()

    for i in range(3):
      print("******** GPU SPEED ********")
      start =time.time()
```

```
[2] x_gpu = x.to(device)
    y_gpu = y.to(device)
    torch.cuda.synchronize()

    for i in range(3):
      print("******** GPU SPEED ********")
      start =time.time()
      result_gpu = torch.matmul(x_gpu,y_gpu)
      torch.cuda.synchronize()
      print(time.time() - start)
      print("verify device : ", result_gpu.device)
```

Another Example: Using Pytorch



- **CPU Specs:**
  - Intel ® i7 – 8550 @ 3.9 GHz
  - 4 Cores (8 logical processors)
- **GPU Specs:**
  - Tesla T4 16GB
  - 2560 CUDA Cores
  - 320 NVIDIA Tensor Cores

Summary

- Accelerator based computing is trending
- GPU is a famous example of an accelerator
- CUDA is the Nvidia's way of programming GPUs
- CPU controls everything

Next Lecture

- More CUDA programming details
  - Threads
  - Blocks
  - Synchronization
  - examples